

Sarah Djebali - PhD
 INSERM CRHC Researcher (HDR)
 IRSD INSERM U1220 - Bat B - CHU Purpan
 31024 Toulouse Cedex 3
 Email: sarah.djebali@inserm.fr
 Nationality: French
 Born: January 15th 1978 (48 years-old)
 Married, 2 children

**Bioinformatics researcher expert in functional
genomic data analysis and method development**

I Training

- | | |
|-----------|---|
| 2002-2006 | PhD in bioinformatics at the IBENS (Institute of Biology of Ecole Normale Supérieure) and the IBISC (Computer science, Integrative Biologie and Complex systems) laboratories
<i>A formal framework for expert eukaryotic gene annotation</i>
With honors (<i>mention très honorable</i>) |
| 2001-2002 | Research Master of Application of Mathematics and Computer science to Biology (AMIB) from Evry University (now Paris Saclay)
<i>Automatic generation of Tagcc programs by training</i>
With honors (<i>mention bien</i>) |
| 1998-2001 | Master in Computer science (software engineer) from ENSEIRB (Ecole Nationale Supérieure de Electronics, Computer science and Radiocommunications of Bordeaux)
With honors (<i>mention bien</i>) |

II Professional career

- | | |
|-----------|---|
| From 2019 | INSERM permanent researcher (CRCN) , Toulouse, France
IRSD (Institute of Research in Digestive Health), INSERM U1220
<i>Complex gene expression regulation to better understand human diseases</i> |
| 2018-2019 | INRAE Senior Postdoc , Toulouse, France
GenPhySE (Genetics, Physiology and Livestock Systems), INRAE U1289
<i>Robust identification of structural variants in livestock species</i> |
| 2015-2018 | Agreenskills+ INRAE Senior Postdoc , Toulouse, France
GenPhySE (Genetics, Physiology and Livestock Systems), INRAE U1289
<i>Functional and comparative annotations of the genomes of four vertebrate livestock species</i> |
| 2010-2015 | CRG Staff Scientist , Barcelona, Spain
CRG (Center for Genomic Regulation)
<i>Coordination of the transcriptomic data analysis of the international ENCODE and MouseENCODE projects</i> |

2006-2010	CRG/IMIM junior Postdoc , Barcelona, Spain CRG/IMIM (Institute of Medical Research) Supervisor: Roderic Guigó <i>Characterisation of the human transcriptional landscape (ENCODE)</i>
2002-2006	PhD / Assistant Professor at Evry University , Evry, France IBISC and IBENS laboratories PhD supervisors: Franck Delaplace and Hugues Roest Crollius <i>A formal framework for expert eukaryotic gene annotation</i>
2002 (6 months)	AMIB Master internship , Evry, France IBISC laboratory Supervisor: Franck Delaplace <i>Automatic generation of Tagcc programs by training</i>
2001 (10 months)	ENSEIRB project graduation , Sheffield, U.K. Sheffield Hallam University Supervisor: Barrie W. Jervis <i>Neural network development for electroencephalogram data classification</i>
2000 (3 months)	ENSEIRB second year project , Bordeaux, France LABRI (Laboratory for research in computer science) Supervisor: David Sherman <i>Inductive databases for the study of gene neighborhood</i>

III Teaching

The following table provides a summary of my teaching activities in the reverse chronological order:

Years	Number of hours	Subject	Audience	Location
2025	4h	Functional annotation databases and their interrogation	Bioinfo Purpan	Toulouse
2024	2h	Functional annotation databases and their interrogation	INSERM workshop attendees	Bordeaux
2024	2h	Causal variant fine-mapping using functional annotations	INSERM workshop attendees	Bordeaux
2022	12h	Linux and the command line	Bioinformatics Master	UPS, Toulouse
2021-2022	4h	Functionnal annotation and common diseases	Bioengineering Master	UPS, Toulouse
2012-2020	72h	Theoretical and practical course about ENCODE and/or the analysis of RNA-seq data	Researchers and engineers	Pasteur/EMBO/CRG/SIB
2011	8h	Theoretical and practical computer science courses (UNIX and programming)	Researchers in Biology	CRG
2002-2006	280h	Theoretical and practical computer science and bioinformatics courses	License and Master students	Evry University

IV Research

IV.1 Publications

I have co-authored **44 original articles** in peer-reviewed journals, of which:

- **7** as first author or first co-author
- **4** as last author

I have also contributed to:

- 1 review article
- 1 book chapter (as first author)
- 1 popular article (as first author)

IV.2 Oral presentations

I have given 15 oral presentations in conferences, of which:

- 5 for which I was invited
- 10 in which my work has been selected by peer-review

IV.3 H-index and citations

According to Research Gate (April 1st 2026), my h-index is **29** and my number of citations is **32,311**, with *Landscape of transcription in human cells* cited **5,022** times. Three of my articles have also been commented by other researchers.

IV.4 Prizes

I have obtained the *Vanguardia de la Ciència* prize for my contribution to *Landscape of transcription in human cells*, *Nature*, 2012.

V Supervision

The following table indicates the people I have scientifically supervised or co-supervised by reverse chronological order:

Person's name	Project title	Level	Supervision ratio	Supervision dates
Victor Lefebvre	Divergence of duplicated genes' cis-regulatory landscape	M2 and PhD student	33%	From January 2024
Thao-Ahn Hoang	The promoter capture hic and the h3k27ac hichip techniques to identify enhancer/gene relationships	M1	100%	October to December 2025
Amel Benarbia	Characterizing genomic rearrangements at transcribed loci following double stranded breaks	M2	50%	January to July 2025
Mathis Lafitte	Differential analysis of mirnas from array experiments	M1	100%	June 2025
Juliette Caverzan	Implementing the score 2 of cardiovascular risk in the next 10 years	L2	100%	May 2025
Océane Carpentier	Knowledge graphs to identify miRNA/gene relationships	M2	50%	January to July 2024
Martin Racoupeau	Comparison of methods for microRNA identification	M2	50%	January to July 2023
Zoé Gerber	Identification of causal variants in coronary artery disease	M2	100%	September 2022 to June 2023
Tess Azevedo	Gene co-expression networks and module identification in livestock species	M1	33%	June to October 2022
Smahane Chalabi	Impact of the maternal diet's fiber content on the transcriptomes and epigenomes of fetus and piglets	Postdoc	50%	October 2021 to March 2023
Tristan Hoellinger	Genome-wide identification of enhancer/gene relationships	Master	100%	September 2020 to February 2022
Pierre Guenzi-Tibéri	Development of a Nextflow pipeline for quantifying the expression of small RNAs from high-throughput sequencing (HTS) data	M1	50%	June to August 2021
Claire Martin	Combining enhancer/gene relationships and genetic data to identify new variants and genes associated to Parkinson's disease	Engineer	50%	September 2020 to March 2021
Camille Mestre	Genome-wide identification of enhancer/gene relationships	M2	60%	January to July 2018
Matthew Smart	Comparison of chromatin accessibility and chromatin structure data	L3	70%	March to September 2017
Nikki Gentle	Transcriptional Signatures of Viral Control in HIV-1 Infected South African Women	PhD student	30%	January to July 2014
Alessandra Breschi	Human and mouse transcriptome comparison	PhD student	50%	November 2012 to October 2015
Bernardo Rodríguez Martín	Chimeric transcript identification in vertebrate genomes	M2 and PhD student	100% and then 30%	January 2013 to October 2016
Emilio Palumbo	Robust pipeline development for the analysis of HTS functional data	Engineer	80%	November 2012 to October 2015
Matthieu Muffato	Development of a user interface for the Exogean gene identification program	M1	80%	April to July 2006
Charles Vejnar	Development of a pseudogene detection module for the Exogean gene identification program	M1	80%	April to July 2005
Khaled Yahiaoui	Improvement and evaluation of Exogean , a eukaryotic gene identification program	M2	80%	January to July 2005

VI Cooperation programs

The following table lists the cooperation programs I have been involved in and my role in them:

Funding source (Country)	Program's name	Program's goal	My role	Program's dates
CNRS MITI (France)	EvoChroCoDeeL	Evolution of regulatory chromatin contacts: insights from massive genomic data and deep learning	WP leader	2025-2026
H2020 (Europe)	GENE-SWitCH	The regulatory GENomE of SWine and CHicken: functional annotation during development	Co-WP leader	2019-2023
INRAE Animal Genetic's division (France)	Raffine	RNA-seq and ATAC-seq for fine-mapping QTL related to pig meat quality	Coordinator	2017-2018
Agreenskills+ (Europe)	FR-AgENCODE data analysis	Multi-species annotation of transcriptome and chromatin structure in domesticated animals	Co-coordinator	2015-2018
NHGRI (USA)	ENCODE3 transcriptome	Investigating the transcriptome of primary cells	Co-coordinator of the data analysis	2011-2015
NHGRI (USA)	MouseENCODE	Producing an Encyclopedia of mouse functional elements	Co-coordinator of the transcriptomic data analysis	2008-2012
NHGRI (USA)	ENCODE2 Data Analysis Center	Coordinating and assisting the integrative analysis of HTS functional data produced by the ENCODE consortium	Co-coordinator	2008-2012
NHGRI (USA)	ENCODE2 Transcriptome	The first exhaustive characterization of the human transcriptome landscape	Co-coordinator of the data analysis	2008-2012